

HARTFORD-BRAINARD, CT, USA

WMO: 725087

Lat: 41.736N

Lon: 72.651W

Elev: 19

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14752

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.3	11.8	-8.9	3.4	11.9	-4.5	4.3	15.3	22.6	32.8	20.2	28.1	7.0	320	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	91.2	73.4	88.5	72.2	85.8	70.9	76.9	86.2	75.4	83.6	74.0	81.2	8.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.1	127.3	81.0	72.8	121.8	79.6	71.8	117.7	78.6	40.2	86.0	38.8	84.0	37.5	80.8	90.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.0	17.1	15.4	DB	0.7	96.2	4.8	3.4	-2.8	98.7	-5.6	100.7	-8.3	102.6	-11.8	105.0	(4)
(5)			WB	-0.5	79.5	4.5	1.8	-3.7	80.8	-6.4	81.9	-8.9	82.9	-12.2	84.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	52.5	29.3	31.4	38.9	50.4	60.7	69.4	75.1	73.6	66.4	54.7	43.8	34.4	(6)
(7)		DBStd	17.80	10.09	9.21	8.87	8.00	7.22	6.35	5.14	5.30	6.75	7.65	8.13	8.64	(7)
(8)		HDD50	2365	643	522	360	88	3	0	0	0	0	39	219	491	(8)
(9)		HDD65	5491	1106	940	810	445	177	28	1	3	63	331	637	950	(9)
(10)		CDD50	3263	2	2	16	99	334	583	779	733	492	184	33	6	(10)
(11)		CDD65	912	0	0	0	6	43	161	315	271	106	10	0	0	(11)
(12)		CDH74	7525	0	0	7	86	451	1344	2737	2138	699	62	1	0	(12)
(13)		CDH80	2509	0	0	1	27	141	446	1012	703	171	8	0	0	(13)
(14)	Wind	WSAvg	6.5	6.9	7.2	7.9	7.6	6.7	6.2	5.8	5.3	5.5	6.0	6.4	6.4	(14)
(15)	Precipitation	PrecAvg	43.1	3.2	3.1	3.5	3.8	3.9	3.8	3.7	3.4	3.8	3.3	3.7	3.7	(15)
(16)		PrecMax	57.8	12.8	8.7	8.2	10.9	11.1	11.9	8.0	9.3	9.8	7.6	7.9	7.8	(16)
(17)		PrecMin	28.4	0.4	0.4	0.6	1.2	0.6	1.5	0.7	0.3	1.5	0.4	0.5	0.6	(17)
(18)		PrecStd	8.3	2.6	1.7	1.8	2.0	2.2	2.5	1.8	2.3	2.4	1.7	1.9	2.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.4	60.8	72.8	83.8	90.0	92.7	96.6	94.4	89.7	80.5	69.9	63.6	(19)
(20)			MCWB	54.5	51.2	56.8	63.8	68.7	74.3	76.5	75.9	73.6	66.1	60.9	56.7	(20)
(21)		2%	DB	54.4	54.1	63.5	74.6	83.8	88.8	91.9	89.9	84.3	74.7	64.4	56.9	(21)
(22)			MCWB	49.2	46.4	51.8	58.0	66.6	72.0	73.9	73.7	69.9	65.2	58.0	52.5	(22)
(23)		5%	DB	48.0	48.8	57.2	69.2	79.2	85.1	88.8	86.7	81.2	71.3	61.4	52.5	(23)
(24)			MCWB	42.9	42.1	47.7	54.9	63.8	69.8	72.6	72.0	68.9	63.4	54.8	48.3	(24)
(25)		10%	DB	43.1	44.7	52.7	64.1	74.1	81.8	85.7	83.8	78.0	67.6	57.4	47.6	(25)
(26)			MCWB	37.9	38.4	44.9	52.4	61.5	68.3	71.6	70.4	67.3	60.7	51.3	42.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.6	55.1	59.8	65.9	72.3	76.9	79.7	78.6	75.7	71.4	64.2	59.0	(27)
(28)			MDCB	59.9	57.1	70.1	80.2	85.0	88.6	89.9	89.4	83.5	75.7	67.0	62.1	(28)
(29)		2%	WB	50.3	47.9	54.3	62.1	69.3	74.7	77.2	76.7	73.7	67.9	60.0	53.8	(29)
(30)			MDCB	52.9	52.0	60.2	71.4	78.7	84.6	86.8	85.0	80.0	72.2	63.3	56.2	(30)
(31)		5%	WB	43.6	42.4	49.6	58.1	66.7	72.9	75.7	75.1	72.1	64.7	56.6	48.4	(31)
(32)			MDCB	48.7	47.3	55.7	64.9	75.9	81.2	84.1	82.3	78.0	69.8	60.4	52.1	(32)
(33)		10%	WB	38.7	39.0	45.7	54.4	64.0	70.8	74.2	73.4	70.0	61.6	52.4	43.4	(33)
(34)			MDCB	42.3	44.2	51.5	61.6	71.3	78.1	81.6	79.6	75.5	66.7	56.6	46.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	14.6	15.8	17.1	19.3	19.6	18.5	18.0	17.7	18.4	18.0	16.2	14.0	(35)
(36)			MCDBR	19.8	21.6	24.8	27.6	26.5	23.2	21.1	20.3	20.9	22.0	21.3	19.9	(36)
(37)			MCWBR	16.2	15.5	15.2	14.3	12.5	10.1	7.5	7.8	9.6	12.9	15.6	17.5	(37)
(38)		5% WB	MCDBR	19.3	19.8	22.0	23.2	22.6	19.9	17.5	16.9	16.5	18.9	18.8	19.0	(38)
(39)			MCWBR	16.8	15.9	16.0	14.3	12.3	10.4	7.8	8.0	9.5	13.5	16.8	18.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.303	0.315	0.339	0.374	0.418	0.463	0.484	0.452	0.380	0.345	0.316	0.301	(40)	
(41)		taud	2.448	2.424	2.428	2.366	2.269	2.181	2.133	2.214	2.389	2.500	2.539	2.530	(41)	
(42)		Ebn at Noon	269	283	289	284	272	260	252	257	271	269	264	262	(42)	
(43)		Edh at Noon	24	29	33	37	42	46	48	43	34	27	22	21	(43)	
(44)	All-Sky Solar Radiation	RadAvg	582	854	1168	1508	1682	1826	1878	1670	1375	911	637	479	(44)	
(45)		RadStd	46	76	88	163	172	180	107	93	108	89	52	46	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (80 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+184	+95	

Nomenclature: See separate page